

(Time: 3 Hours)

- N.B:** 1) Question **number 1** is compulsory.
2) Attempt **any THREE** out of the remaining.
3) Assume suitable data if **necessary** and justify the assumptions.
4) Figures to the **right** indicate full marks.

Q 1

- A** Explain key Features of Global Scheduling algorithm. [05]
B What are the services offered by middleware? [05]
C Differentiate between RMI and RPC. [05]
D Explain synchronization in DFS with its challenges. [05]

Q 2

- A** Justify how load balancing is useful in distributed system. [10]
B What is fault tolerance? Explain failure models. [10]

Q 3

- A** Explain group communication. [10]
B Explain Raymond's tree based algorithm for mutual exclusion. [10]

Q 4

- A** Explain Ricart-Agrawala's algorithm and how it optimizes the message overhead in achieving mutual exclusion. [10]
B Discuss Google file system (GFS) as a scalable distributed file system. [10]

Q 5

- A** Explain bully election algorithm. [10]
B Discuss code migration in distributed system. [10]

Q 6

- A** Explain any five data centric consistency models. [10]
B Discuss design issues in distributed systems. [10]
